

Research Methodology

Concept of research design and its importance, features of a good research design, exploratory research design, descriptive research design, experimental research design, quantitative & qualitative research, empirical research, case study, applications of different research designs and their limitations, concept of dependent variable, independent variable and assessment of their relationships, issue of generalization of research findings

Unit 2: Research Design (6 Hrs.)

Concept of Research Design:

Design means a plan, drawing or blueprint. It produces to show the looks and function or workings of building or something else. It is the preparation of the design of the research project, popularly known as the “research design”. Decisions regarding what, where, when, how much, by what means concerning an inquiry or a research study constitute a research design. A research design is the arrangement of conditions for collection and analysis of data in a manner that aims to combine relevance to the research purpose with economy in procedure. The research design is the conceptual structure within which research is conducted. It constitutes the blueprint for the collection, measurement and analysis of data. As such the design includes an outline of what the researcher will do from writing the hypothesis and its operational implications to the final analysis of data. More explicitly, the designs decisions happen to be in respect of:

- What is the study about?
- Why is the study being made?
- Where will the study be carried out?
- What type of data is required?
- Where can the required data be found?
- What periods of time will the study include?
- What will be the sample design?
- What techniques of data collection will be used?
- How will the data be analyzed?
- In what style will the report be prepared?

According to Willian Zikmund, Research design is defined as a master plan specifying the methods and procedure for collecting and analyzing the needed information. It helps to decide upon issues like what, when, where. How much by what means etc. with regards to research study.

Location where study will be conducted, nature of data required, sample design used, time period needed, from where data will be collected, technique of data collection, method of data analysis, type of report preparation.

Concept of Research Design:

- (i) Research design is a logical and systematic outline of research project prepared for directing, guiding, and controlling a research work.
- (ii) It means to prepare detailed plan and procedure for the conduct of the research project.
- (iii) It acts as a broad outline of the research work and acts as a master plan/blueprint for the conduct of formal investigation.
- (iv) It is the basic plan that guides researcher in the execution of research project under taken.

(v) Before the commencement of work of research project, the researcher has to prepare research design.

Elements of research design:

- (i) Nature of research and objective study.
- (ii) Time period of research study.
- (iii) Universe and sample size of respondents.
- (iv) The location where the study would be conducted.
- (v) The research required to conduct the research.
- (vi) Type and source of research data required.
- (vii) Techniques of data collection and analysis.

Question 1: What do you mean by research design?

Need and Importance of Research Design:

1. Provides guidelines

Research design provides guidelines to the researcher in respect of

- (i) Time period of research work
- (ii) Type and source of data to be collected
- (iii) Technique of data collection and analysis
- (iv) Sampling design

Therefore, the researcher will be able to collect right data from the right source at the right time.

2. Organizing resources:

Research design enables organizing of resource required to carry on research activity. The resources required are:

- (i) Funds
- (ii) Equipment/instrument and materials
- (iii) Manpower

Availability of right amount of resources facilitates smooth conduct of research activity.

3. Selection of techniques of data collection and analysis:

Research design helps to select appropriate techniques for data collection such as

- (i) Survey or interview
- (ii) Observation

(iii) Experimentation

(iv) Internet

(v) Library etc.

Research design also helps to select approximate technique for data analysis such as:

(i) Measure of central tendency (mean, median, mode)

(ii) Time series (Simple moving average, weighted moving average)

(iii) Collection and representation technique etc.

4. Collection of relevant data:

The research design indicates sources as

(i) Area of research

(ii) Universe/population of research

(iii) Sample size etc.

According to researcher can select right area of research and target audience. He can decide universe/population from which samples are selected to collect the relevant data.

5. Objective of research:

Research design specifies objective of research. Research design provides right direction of the researcher to carry on with research activity. This in turn will help to attend research objectives.

6. Monitoring of expenditure:

(i) Research design includes allocation of research budget for various research activities.

(ii) There is proper control over expenditure.

(iii) Waste of funds does not happen.

(iv) The research is successfully conducted with allocation funds.

7. Execution of research work:

The research design indicates the start time and completion time of research activity. Therefore, there is timely execution of research work. If research design does not indicate the time frame, there is every possibility of research work getting delayed and objectives may not be achieved.

8. Motivation of research staff:

A systematic research design motivates the staff to collect the right data from the right source. Also due to timely completion of research activity, the research staff may be rewarded with monetary and non-monetary incentives.

9. Improvement of decision making:

- (i) Research design enables researcher to collect right data from right source.
- (ii) Right data enables to take right decision.
- (iii) Wrong data may lead to misleading decision making.

Question 2: What are the important of research design?

Features of good research design:

A good design is often characterized by adjectives like flexible, appropriate, efficient, and economical and so on. Generally, the design which minimizes bias and maximizes the reliability of the data collected and analyzed is considered a good design. The design which gives the smallest experimental error is supposed to be the best design in many investigations. Similarly, a design which yields maximal information and provides an opportunity for considering many different aspects of a problem is considered most appropriate and efficient design in respect of many research problems. Thus, the question of good design is related to the purpose or objective of the research problem and also with the nature of the problem to be studied.

A research design appropriate for a particular research problem, usually involves the consideration of the following factors:

- (i) The means of obtaining information
- (ii) The availability and skills of the researcher and his staff, if any
- (iii) The objective of the problem to be studied;
- (iv) The nature of the problem to be studied; and
- (v) The availability of time and money for the research work.

It is also features of good research design are:

1. Good research design must be realistic, workable, appropriate and able to give us intended information.
2. Design must be flexible, efficient, thrift and parsimonious to economy.
3. Design should be consistent with the research capability of the researcher or feasible for the research.
4. Research design must be based and synchronize with the purpose of the research process.
5. It must be flexible, so that it can be changed as per the situation change.
6. Good design should be formulated after the crucial study of the problem.
7. Design should provide well developed guidelines for entire research steps.
8. Design should constitute valid, reliable and generalizable features.
9. Research design should over the data collection and analysis technique properly.
10. Design should able to recommended appropriate methods for hypothesis formulation and testing.

Features of good research design:

1. Objectivity
2. Reliability
3. Validity
4. Sufficient information
5. Other features

Objectivity: Objectivity refers to the ability of the research instruments to give conclusions that are free from observer's personal biases. A good research design should be able to select those instruments only that provide objective conclusions. Usually, it is believed that maintaining objectivity is pretty easy, but it proves to be difficult during execution of research and data analysis.

Reliability: Another essential feature of a good research design is the reliability of responses. The instruments used in research should be able to provide similar responses to a question asked from a respondent. If the response varies, the instrument is considered unreliable. In other words, reliability of research design is measured in terms of consistency in responses.

Validity: An important characteristic of a good research design is its ability to answer the questions in the way it was intended to. It should focus on the objective of the research and make specific arrangements or plan for achieving that objective.

Generalisability: A research design is said to be generalisable if the outcome of the research is applicable on a bigger population from which the sample is selected. A research design can be made generalisable by properly defining the population properly, selecting the sample carefully, analyzing the statistical data appropriately, and by preparing its methodology. Therefore, the more the outcomes are generalisable, more efficient is the research design.

Sufficient Information: Any research is conducted to gain insight of the hidden facts, figures and information. The research design should be able to provide sufficient information to the researcher so that he can analyze the research problem in a broad perspective. The research design should be able to identify the research problem and research objective.

Other Features: Along with the above, there are some other features also that make a research design good. These are adaptability, flexibility, efficiency, etc. A good research design should be able to minimize the errors and maximize the accuracy.

Question 3: Discuss the characteristics of research design.

Exploratory Research Design:

- Exploratory research design is also termed as formulating research studies.
- It is a research conducted for a problem that has not been clearly defined.
- In this type of design, researchers try to uncover the hidden and unknown facts and phenomena.
- Exploratory research, as the name states, intends merely to explore the research question and does not intend to offer final and conclusive solutions to existing problems.
- It is conducted to determine the nature of the problem; this type of research is not intended to provide conclusive evidence but helps us to have a better understanding of the problem.
- Major focus of this design is to gain knowledge.
- Discover new data.

- It is also known as formulative research.

Example (1): Suppose a juice bar owner feels that increasing the variety of juices will enable increase in customers, however he is not sure and needs more information.

The owner intends to carry out an exploratory research to find out if expanding their juices selection will enable him to get more customers or there is any other better idea.

Example (2): Service such as Google alerts may send major search engine, search results by researcher via e-mail. Services such as Google tends track. Google comprehensive search results over lengthy periods of time. Researcher may set up website to attract world-wide feedback on any subject.

Types of Exploratory Research Design:

Primary research method:

- Primary research is information gathered directly from the subject.
- Projective techniques, Depth interview, experience survey, Focus group, Observation.

Secondary research method:

- Secondary research is gathering information from previously published primary research.
- In such research you gather information from sources like case studies, magazines, newspapers, books etc.

Major types of qualitative techniques for exploratory research design:

1. Projective technique:

- Under this technique, individuals are asked to express their opinions and feelings on a situation or behavior another individual.
- Hypothetical situation/behavioral patterns to individuals are selected, which are ambiguous and vague in nature, and respondents are requested to express their beliefs and feelings in the same.
- In doing so, researchers are able to study the motivations and attitudes of the respondents themselves, thereby, enabling them to dig deeper into the research.

Types of projective techniques

(a) Sentence completion:

Under this technique, participants are provided with incomplete sentence that are referred to as 'selems' and encouraged to complete these sentences their own way. Example: "Taking a holiday in the mountain is"

(b) Word Association:

Under this technique, participants are provided with a list of words like village, cities, one by one, and requested to state the first thing that comes to their mind in response to each of these words.

(c) Story Completion:

Under this technique, participants are given an incomplete story which, though subtly leads to a predetermined topic, does not in any provide an insight into the ending. Participants are required to conclude the story in their own way.

(d) Verbal Projection Tests or Thematic Apperception Tests:

In these tests, a situation is described to the participants either orally, or by exploiting simple methods of visual presentation. Participants are then required to provide their thoughts and opinions on the effects the situation could have on others, not themselves.

2. Depth Interview:

Small group of respondents is intensively interviewed on individual basis so as to explore their perception about a particular event, concept, idea or situation, it is called depth interview.

Example: A depth interview can be conducted to explore the reviews of participants of an organizational event about the nature of operations, activities, its usefulness or changes required if any. In depth interviews are often quite long and can involve multiple interviews with one participant.

Numerous topics or issues like strategic planning, program improvement, problems identification, need analysis etc.

3. Experience Survey:

Experts or individuals having significant information on the concerned topic are approached for collecting relevant information, it is called Experience Survey. This is the reason experience surveys are also called expert opinion surveys. The only purpose here is to utilize the experience of these experts in gaining valuable insight of the problem.

A customer experience survey is a questionnaire designed to help a business captures customers thoughts and feelings about its brand, product or services.

Example: In order to deal with a consumer product problem, a marketing executive would be sufficient to provide required information as he is having experience in the field of consumers marketing. The information provided by him may be very useful for the research problem. Information can also be taken from sales executives, sales manager, product manager or any senior officer of the organization.

4. Focus Groups:

Focus group is a method in which a group of respondents is selected so as to collect desired information through a formal interactive session. It is also called group interviewing method. In this, the respondents response to the questions of a moderator (sometimes researcher himself/herself) to discuss about the given topic or research problems, moderator asks specially designed questions so as to explore in depth information. The information explored here is not possible to collect from other techniques like surveys, observations or interviews.

Focus group method is characterized by the use of effective. It significant method and strategies so as to collect and or the analysis the information generated here. This makes focus group method a reliable source of data collection. This type of method is used in areas like vital new product concept or new product development, improvement of production line in an organization, motivation program for few employees etc.

5. Observation:

The method involving the systematic recording and assessment of behavioral forms of a particular individual (or group of individual), object or event so as to collect desired information about a research problem without any personal communication is called observation. Here, the observer observes the object from a suitable distance or through audio video recordings. The information is recorded for future use. The respondents may be observed in their natural environment or a contrived environment is designed for the purpose. The approach for observation may be direct or indirect, structured or unstructured. The observation method is very popular for behavioral science researches.

Steps to conduct a research:

Identify the problem: A researcher identifies the subject of research and the problem is addressed by carrying out multiple methods to answer the questions.

Create the hypothesis: A researcher has found out that there are no prior studies and the problem is not precisely resolved, the researcher will create a hypothesis based on the questions obtained while identifying the problem.

Further research: Once the data has been obtained, the researcher will continue his study through descriptive investigation. Qualitative methods are used to further study the subject in detail and find out if the information is true or not.

Question 4: Define exploratory research design and discuss the major types of qualitative techniques for exploratory research design. Write advantage of exploratory research.

Advantage of Exploratory Research

- The researcher has a lot of flexibility and can adapt to changes as the research progresses.
- It is usually low cost.
- It helps lay the foundation of a research, which can lead to further research.
- It enables the researcher understand at an early stage, if the topic is worth investing the time and resources and if it is worth pursuing.
- It can assist other researchers to find out possible causes for the problem, which can be further studied in detailed to find out, which of them is the most likely cause for the problem.

Disadvantage of Exploratory Research

- Even though it can point you in the right direction towards what is the answer it is usually inconclusive.
- The main disadvantage of exploratory research is that they provide qualitative data. Interpretation of such information can be judgmental and biased.
- Most of the times, exploratory research involves a smaller sample, hence the results cannot be accurately interpreted for a generalized population.
- Many times, if the data is being collected through secondary research, then there is a chance of that data being old and is not updated.

Descriptive Research Design

- Descriptive research aims to accurately and systematically describe a population, situation or phenomenon.
- It can answer what, where, when and how questions, but not why questions.
- It can use a wide variety of research methods to investigate one or more variables.
- Unlike in experimental research, the researcher does not control or manipulate any of the variables, but only observes and measures them.

Example:

- An apparel brand that wants to understand the fashion purchasing trends among New York buyers will conduct a demographic survey of this region, gather population data and then conduct descriptive research on this demographic segment. The study will then uncover details on “what is the purchasing pattern of New York buyers,” but not cover any investigative information about “why the pattern exists. Because of the apparel brand trying to break into this market, understanding the nature of their market is the study’s objective.

- How much time does an average teenager spend on watching documentaries on OTT platforms?
- From where customer purchase their FMCG Goods are Mall, mom and pop shop, super market?

Types of Descriptive Research Design

1. Longitudinal Design
2. Cross-sectional Design

Longitudinal Design:

A longitudinal study is a research conducted over an extended period of time. It is mostly used in medical research and other areas like psychology or sociology. A longitudinal study, a type of correlation research on a number of variables without drawing to enrolled them and this type of study can take place weeks, months or even years.

Cross-sectional Design:

Data is collected from a population at a specific point in time. A study can be done in which data gathered just once, perhaps over a period of days, weeks, or months, in order to answer a research question is called one shot or cross sectional studies.

Question 5: Define descriptive research design. Describe the types of descriptive research design.

Characteristics of Descriptive Research Design

- While descriptive research can employ a number of variables, only one variable is required to conduct a descriptive study.
- Descriptive studies are closely associated with observational studies, but they are not limited with **observation** data collection method. Case studies and **surveys** can also be specified as popular data collection methods used with descriptive studies.
- Findings of descriptive researches create a scope for further research. When a descriptive study answers to the question “What?”, a further research can be conducted to find an answer to “Why?” question.

Advantage of Descriptive Research Design:

- Effective to analyses non-quantified topics and issues
- The possibility to observe the phenomenon in a completely natural and unchanged natural environment
- The opportunity to integrate the qualitative and quantitative methods of data collection. Accordingly, research findings can be comprehensive.
- Less time-consuming than quantitative experiments
- Practical use of research findings for decision-making

Disadvantage of Descriptive Research Design:

- Descriptive studies cannot test or verify the research problem statistically
- Research results may reflect certain level of bias due to the absence of statistical tests
- The majority of descriptive studies are not ‘repeatable’ due to their observational nature
- Descriptive studies are not helpful in identifying cause behind described phenomenon

Question 6: Describe the characteristics, advantage and disadvantage of descriptive statistics.

Experimental research design

Experimental research design is a framework of protocols and procedures created to conduct experimental research with a scientific approach using two sets of variables. Herein, the first set of variables acts as a constant, used to measure the differences of the second set. The best example of experimental research methods is [quantitative research](#).

Experimental research helps a researcher gather the necessary data for making better research decisions and determining the facts of a research study.

A researcher can conduct experimental research in the following situations —

1. When time is an important factor in establishing a relationship between the cause and effect.
2. When there is an invariable or never-changing behavior between the cause and effect.
3. Finally, when the researcher wishes to understand the importance of the cause and effect.

Importance of Experimental Research Design

To publish significant results, choosing a quality research design forms the foundation to build the research study. Moreover, effective research design helps establish quality decision-making procedures, structures the research to lead to easier data analysis, and addresses the main research question. Therefore, it is essential to cater undivided attention and time to create an experimental research design before beginning the practical experiment.

By creating a research design, a researcher is also giving oneself time to organize the research, set up relevant boundaries for the study, and increase the reliability of the results. Through all these efforts, one could also avoid inconclusive results. If any part of the research design is flawed, it will reflect on the quality of the results derived.

Types of Experimental Research Designs

Based on the methods used to collect data in experimental studies, the experimental research designs are of three primary types:

1. Pre-experimental Research Design

A research study could conduct pre-experimental research design when a group or many groups are under observation after implementing factors of cause and effect of the research. The pre-experimental design will help researchers understand whether further investigation is necessary for the groups under observation.

Pre-experimental research is of three types

1. One-shot Case Study Research Design
2. One-group Pretest-posttest Research Design
3. Static-group Comparison

2. True Experimental Research Design

A true experimental research design relies on [statistical analysis](#) to prove or disprove a researcher's hypothesis. It is one of the most accurate forms of research because it provides specific scientific evidence. Furthermore, out of all the types of experimental designs, only a true experimental design can establish a cause-effect relationship within a group. However, in a true experiment, a researcher must satisfy these three factors —

1. There is a control group that is not subjected to changes and an experimental group that will experience the changed variables
2. A variable that can be manipulated by the researcher
3. Random distribution of the variables

This type of experimental research is commonly observed in the physical sciences.

3. Quasi-experimental Research Design

The word “Quasi” means similarity. A quasi-experimental design is similar to a true experimental design. However, the difference between the two is the assignment of the control group. In this research design, an independent variable is manipulated, but the participants of a group are not randomly assigned. This type of research design is used in field settings where random assignment is either irrelevant or not required.

The classification of the research subjects, conditions, or groups determines the type of research design to be used.

Advantages of Experimental Research

Experimental research allows you to test your idea in a controlled environment before taking the research to clinical trials. Moreover, it provides the best method to test your theory because of the following advantages:

1. Researchers have firm control over variables to obtain results.
2. The subject does not impact the effectiveness of experimental research. Anyone can implement it for research purposes.
3. The results are specific.
4. Post results analysis, research findings from the same dataset can be repurposed for similar research ideas.
5. Researchers can identify the cause and effect of the hypothesis and further analyze this relationship to determine in-depth ideas.
6. Experimental research makes an ideal starting point. The collected data could be used as a foundation to build new research ideas for further studies.

Question 7: Define experimental research design. Write the important of experimental research design. Also discuss the types of experimental research design.

Mistakes to Avoid While Designing Your Research

There is no order to this list, and any one of these issues can seriously compromise the quality of your research. You could refer to the list as a checklist of what to avoid while designing your research.

1. Invalid Theoretical Framework

Usually, researchers miss out on checking if their hypothesis is logical to be tested. If your research design does not have basic assumptions or postulates, then it is fundamentally flawed and you need to rework on your research framework.

2. Inadequate Literature Study

Without a comprehensive [research literature review](#), it is difficult to identify and fill the knowledge and information gaps. Furthermore, you need to clearly state how your research will contribute to the research field, either by adding value to the pertinent literature or challenging previous findings and assumptions.

3. Insufficient or Incorrect Statistical Analysis

Statistical results are one of the most trusted scientific evidence. The ultimate goal of a research experiment is to gain valid and sustainable evidence. Therefore, incorrect statistical analysis could affect the quality of any quantitative research.

4. Undefined Research Problem

This is one of the most basic aspects of research design. The research problem statement must be clear and to do that, you must set the framework for the development of research questions that address the core problems.

5. Research Limitations

Every study has some [type of limitations](#). You should anticipate and incorporate those limitations into your conclusion, as well as the basic research design. Include a statement in your manuscript about any perceived limitations, and how you considered them while designing your experiment and drawing the conclusion.

6. Ethical Implications

The most important yet less talked about topic is the ethical issue. Your research design must include ways to minimize any risk for your participants and also address the research problem or question at hand. If you cannot manage the ethical norms along with your research study, your research objectives and validity could be questioned.

Experimental Research Design Example

In an experimental design, a researcher gathers plant samples and then randomly assigns half the samples to photosynthesize in sunlight and the other half to be kept in a dark box without sunlight, while controlling all the other variables (nutrients, water, soil, etc.)

By comparing their outcomes in biochemical tests, the researcher can confirm that the changes in the plants were due to the sunlight and not the other variables.

Experimental Research Design

- A causal relationship exists when one variable in a data set has a direct influence on another variable.
- Also known as cause and effect relationship.
- In warmer weather caused more sales of AC or Cooler
- A research method used to investigate the interaction between independent and dependent variables.
- It is used to determine a cause-and-effect relationship.
- Research conducted with a scientific approach using two sets of variables.
- Independent variable and dependent variables.

Examples:

- A researcher wants to examine the effect of advertisement on customer demand.
- Then two groups will be selected, one group will expose to Ad experimental group and second group will be control group.

Independent and Dependent variables

- Those variables which get affect the other variables are called independent variables.
- Those variables which get affected by the other variables are called dependent variables
- In a study to determine whether how long a student sleeps affects test scores, the independent variable is the length of time spent sleeping while the dependent variable is the test score.

Concomitant Variables

Concomitant variable is a variable which we observe during the course of our research or statistical analysis, but we cannot control it and it is not the focus of our analysis. It is also called covariate.

Moderating Variables

- A variables that can strengthen, diminish, negate, or otherwise alter the association between independent and dependent variables.
- Moderating variables can also change the direction of this relationship

Extraneous Variables

- An extraneous variable is any factor that can affect an experiment's dependent variables.
- Treatment groups are the sets of participants in a research study that are exposed to some manipulation or intentional change in the independent variable of interest.
- The treatment group (also called the experimental group) receives the treatment whose effect the researcher is interested in.
- The control group receives either no treatment, a standard treatment whose effect is already known.

Quantitative & Qualitative research

Quantitative research: Quantitative research is the process of collecting and analyzing numerical data. It can be used to find patterns and averages, make predictions, test causal relationships, and generalize results to wider populations. A type of research in which the researcher decides what to study; asks specific, narrow questions, collects quantifiable data from participants (a large number of participants), analyzes these numbers using statistics, and conducts the inquiry in an unbiased, objective manner.

Qualitative research: Qualitative research is the process of collection and analysis of non-numerical data like text, video, or audio to understand concepts, opinions, or experiences. It can be used to gather in-depth insights into a problem or generate new ideas for research. This type of research in which the researcher relies on the views of participants; asks board, general questions, collects data consisting largely or words or text from participants, describes and analyzes these words for themes, and conducts the inquiry in a subjective; biased manners.

Question 8: Define quantitative and qualitative research.

Empirical research

Empirical research is research that is based on observation and measurement of phenomena, as directly experienced by the researcher. The data thus gathered may be compared against a theory or hypothesis, but the results are still based on real life experience.

Case Study

Introduction:

Case study is an intensive study of a case, which may be an individual, a family, a social group, an institution, a community or even entire cultural. Robson (1993) has defined case study as “a strategy for doing research which involves an empirical investigation of a particular contemporary phenomenon within a real life context using multiple source of evidence. Yin (1984) defines the case study research as “an empirical enquiry that investigates contemporary phenomenon within its real life context. Rather than using samples to examine a limited number of variables, case study methods involve in-depth, longitudinal examination of a single instant or event”.

To determine the basic causal relationship between the variables, the studies, which go deep into the causes of things by the method of collecting and investigating very minute samples of a huge number of objects in detail, are called the case studies. In a case study, however, a single unit is examined carefully to observe, to detect, to quantify, to measure, to explore, to resolve, to appraisal and comprehend the issues, concerns and relationships (what is there

and how it got there?) in a particular situation. Often the diagnostic research or clinical researches are the case studies.

The method of case study is, however, different from survey method. Survey are usually extensive (wide spreading or having broad-spectrum), in that a few carefully designed questions are asked for a large number of sample. Case studies are intensive (narrow or specific or concentrated), in that a great number of objects are inquired in detailed in a relatively related small number of samples or for a single unit. A unit may be a person, a small group of people, a single band of party or company or a single episode or an event.

In case study, the investigator gathers pertinent data about the present status, past experience and environmental forces that contributes to the individual and behavior of the unit. After analyzing the sequences and interrelationship of these facts he/she conducts a comprehensive study of the unit as it functions. Its aim and goal is to know patterns of a unit and the place of the unit in its surrounding of social setting technical setting.

From the case studies, various facts (results) can be derived by systematic enquiry. These facts or results often put forward many hypotheses, which can be used as the survey questionnaire or the design of controlled experiment.

The results, finding or theories developed through the case studies may perhaps not be generalized for the other similar cases, but may possibly be compared to other cases to develop a new concept about such cases. In this background, many of the case may seem to be similar in nature but in the certain aspect and feature, one may be unique and therefore it is not representative for other cases.

Types of case studies:

1. Exploratory Case Studies.
2. Hypothesis Testing Case Studies.

Exploratory Case Studies:

In exploratory studies, researcher tries to identify the significant causes (variables) of the case and seeks to establish relation between them. The objective of this study is to develop the hypothesis rather than their testing. The researcher collects the facts concerning an event or a case minutely by diagnostic (investigate) way and sets these facts as the hypothesis to test their significance in this further (supplementary) research works.

Hypothesis Testing Case Studies:

It is a study where researcher seeks to test the relationship between different variables. The aim of this study would be validity of the relationship using the data collected by different methods.

From the exploratory study, the relationship between variables is predicted. A researcher seeks to develop a hypothesis between the variables by minute investigation in exploratory studies and which, is tested in this type of study. Researcher tries to collect the facts from the similar type of case situation and/or from results of the exploratory case studies and inquires about to test their significance, reliability and validity. (the validity of relationship or established hypothesis)

Steps in Case Studies:

The steps of case study methods are quite different than the other methods of research. The steps of case study after designing the research follow a similar pattern to the other methods of research. However, in most of the times the study of the cases has to be done without preplanning. For example, incidents occur unintentionally and in all such

cases one should have to study it and make a report. A patient come to the doctor with a new symptoms of illness, doctor has to identify it and to cure it. A case of murder or suicide observed in a society and police has to identify the case what is and what precaution they should take for prevent such types of cases. This all events what we mentioned are the cases that occurs preplanning, however the concerned person should study such cases without pre plan. Such studies are said to be the case studies before designing.

Here, we discuss first the ways (or the steps) of case study for before designing phase [phase-I] and then for the after designing phase [phase-II]. We have already discussed that the before designing phase of the case studies is called the exploratory case studies and the after designing phase as the hypothesis testing case studies.

The main steps in case studies for phase-I are

- a. **Observation of events:** It is done to determine the present situation of the cases, starting with very little prior knowledge or information.
- b. **Collection of past information:** The back ground information or the data from the past is collected to relate with the present information.
- c. **Enquiring or Diagnosing:** Enquiring or diagnosing the current situation from different angle for indentifying, listing and analyzing the possible causes of current situation relevant to the case.
- d. **Hypothesis setting:** By editing and selecting possible causes from the past information and current diagnosis, the researcher has to formulate a tentative statement regarding the cases (we call it the hypothesis). The researcher selects only the information (or data) relevant to the case, which are important and useful for developing hypothesis.

The research inters into phase-II and it has the steps

- e. **Define research questions and set objective:** It is the process of setting the investigable issues and the purpose of doing the case research.
- f. **Design the case study:** The design covers the plan and strategy for the process of case unit selection, sources of data collection, method and techniques of data collection. Designing of the case study is done for getting answer of what is the unit of study, what characteristic to be examined, what relations are trying to be explored and what process will direct the investigation? Design is done based on the results of the analyzed background information.
- g. **Conduct the case study and collect the data:** In this step more information are collected from further diagnosis and enquiry with supplemented procedures applied by the researcher. The information (data) are collected as determined by the design, from every aspect of the subject, intensively and extensively. Interviews, observations, documents are used to collect data.
- h. **Organization and analysis of case evidence:** The case data are then organized after determining the categories and the analysis of the data can be done either qualitatively and quantitatively however the qualitative method is widely used.
- i. **Testing hypothesis:** The hypothesis developed after the phase-I is tested statistically.
- j. **Testing of validity:** The findings of the test procedure and the results obtained from analysis are put for the validity test.

- k. **Reporting:** In this stage the researcher reports his/her finding, conclusions, recommendations and implication etc. and discusses their significance integrating logical argument.
- l. **Action and correction:** Action and corrections are made to see the effect of change of situation.

Sources of Case Data:

The information for the case study may be personal documented diaries, confessions, autobiographies and memories. The sources of case study may be divided into two categories: Personal documents and case History. They are discussed below:

1. Personal Documents:

These include all those materials, which throw light on the feelings, moods and reactions of a person to his social milieu. The diaries usually include reflections of a person regarding other people and things. The same is the case with memories, autobiographies, etc. These tell us a great deal about their authors as well as social situations. "Diary is a perennial and continuous record of personal reflections. It contains the emotional and intellectual responses of the writer to his multiform experiences. The personal correspondences also are free, frank and intimate. Ordinarily, it is considered boorish and uncivilized to read the correspondence of other people. However, after a lapse of sufficient time many personal correspondences are considered suitable for publication. This is so because personal letters reveal a personal and his situation is very interesting. Similarly, memories and autobiographies are extremely useful sources of information in the case study method.

2. Case History:

In the case study method, besides personal documents detailed above, the case history is another important source of information. In this the researcher collects detailed information about the life of a person and talks to him, whenever needed, to collect and verify the information gathered. For example, if we wish to collect information about the childhood of a man; we shall interview parents of that person, his neighbors, playmates and other individuals who watched him as a child or participated with him in various childhood activities. Besides personal interview, the information about a person can be gathered in various other ways. He may be watched while doing his routines or giving a public lecture or engaged in conversation and debate with friends. Moreover, he may be exposed to various temptations to test his spirit. His reactions will tell us all about his moral qualities. Sometimes a person is hypnotized in order to bring to surface hidden memories. In short, all available socio-psycho-medical methods may be used to collect an authentic case history.

The nature of the sources of data depends on the nature of the study topic. Thus, there exist so many other sources of case data. Some of them are as given below:

- The data obtained from exploratory observations.
- Life history of persons/organization/institutes etc.
- Data obtained through interview and observations like primary sources.
- Careful remarks, notes and enquiry of events.
- Conferences, conversations experimental studies etc.
- Secondary sources such as reports, records, newspaper, magazine, books, files, diaries etc.

Question 9: Define and discuss case study method. Describe the different steps of case study approach.

Limitations of case study:

Case study method is a useful technique of studying social material. However, like all other methods, it has own defects. These defects and limitations are as follows:

1. **False Self-confidence:** Researcher may develop a false conviction that he knows all about the subject under study. He thinks that he needs learn nothing more about it. However, this is fallacious in the nature of the case. No one can claim that he knows anything so fully that no new fact will possible be known about it by anybody. Actually, the claim to know all is foreign to a scientific temper.
2. **Small-Unit-Based Conclusions:** Generalization drawn from a single case cannot be applied to all cases in a given population. However deep and penetrating a study may be, its conclusions will not be generally valid if the unit chosen for study is very small. For example, if a researcher studies a single prostitute and concludes that prostitution in an art. The conclusion may have some validity in cases of highly sophisticated prostitutes but will be generally untrue.
3. **Expensive and Time consuming:** Because of exploratory nature and the deep study of a number of units, the cost and time factor is excessive. It may take years to study some units. Of course time and money spent is worthwhile if some great strides in knowledge become possible, but usually much effort is exhausted in routine things.
4. **Possibility of Error:** There are high chances of making errors in the selection of case, observation, noting and interview. Since the concentration is on a single subject one is less likely to become aware of the mistake and the mistake may become compounded in the absence of countervailing checks. Similarly personal biasness, hunches also may influence on the study. Therefore, case study method is error-prone and only a judicious and cautions researcher can avoid these.
5. **Other limitations:** Besides the above- mentioned limitations, different social scientists have pointed out many other drawbacks. The other drawbacks of the case study method are:
 - The researcher in case study deals with a single case over a long time, so that he/she is liable to develop deep support and sympathy with his subject and may conclude emotionally as his/her wishes.
 - The imagination of the researcher may kill the true narration of events, if the study is done alone rather than cooperative efforts.
 - Case study method is not highly appreciated for well structured phenomena.
 - Case study method does not suit if important and essential variables are clearly identified.
 - If cause and effect variables are already stated, the case study method of research is not appropriate.
 - It is highly useful only for exploratory researches.
 - These studies may terminate to lack of utility when attempting to determine precise relationship between variables.
 - The case study method may be affected by personal biases and preconceived notions.

Despite the above stated limitations, case studies are being undertaken in several disciplines, as a tool of scientific research in view of the several advantages it has. Most of the limitations can be removed if researcher is always conscious of these and are well trained in the modern methods of collecting case data in the scientific techniques of assembling, classifying and processing the same.

Question 10: Explain the limitation of case study.

Dependent and independent variables: A concept which can take on different quantitative values is called a variable. As such the concepts like weight, height, income are all examples of variables. Qualitative phenomena (or the attributes) are also quantified on the basis of the presence or absence of the concerning attribute(s). Phenomena which can take on quantitatively different values even in decimal points are called 'continuous variables'.* But all variables are not continuous. If they can only be expressed in integer values, they are non-continuous variables or in

statistical language ‘discrete variables’. Age is an example of continuous variable, but the number of children is an example of non-continuous variable. If one variable depends upon or is a consequence of the other variable, it is termed as a dependent variable, and the variable that is antecedent to the dependent variable is termed as an independent variable. For instance, if we say that height depends upon age, then height is a dependent variable and age is an independent variable. Further, if in addition to being dependent upon age, height also depends upon the individual’s sex, then height is a dependent variable and age and sex are independent variables. Similarly, readymade films and lectures are examples of independent variables, whereas behavioural changes, occurring as a result of the environmental manipulations, are examples of dependent variables.

Generalization: It is the degree to which you can apply the results of your study to a broader context. Research results are considered generalizable when the findings can be applied to most contexts, most people, most of the time.

Question 11: Define dependent and independent variables.

Limitations Research Design:

It is for sure that your research will have some limitations and it is normal. However, it is critically important for you to be striving to minimize the range of scope of limitations throughout the research process. Also, you need to provide the acknowledgement of your research limitations in conclusions chapter honestly.

It is always better to identify and acknowledge shortcomings of your work, rather than to leave them pointed out to you by your dissertation assessor. While discussing your research limitations, don’t just provide the list and description of shortcomings of your work. It is also important for you to explain how these limitations have impacted your research findings.

Your research may have multiple limitations, but you need to discuss only those limitations that directly relate to your research problems. For example, if conducting a meta-analysis of the secondary data has not been stated as your research objective, no need to mention it as your research limitation.

Research limitations in a typical dissertation may relate to the following points:

1. Formulation of research aims and objectives. You might have formulated research aims and objectives too broadly. You can specify in which ways the formulation of research aims and objectives could be narrowed so that the level of focus of the study could be increased.

2. Implementation of data collection method. Because you do not have an extensive experience in primary data collection (otherwise you would not be reading this book), there is a great chance that the nature of implementation of data collection method is flawed.

3. Sample size. Sample size depends on the nature of the research problem. If sample size is too small, statistical tests would not be able to identify significant relationships within data set. You can state that basing your study in larger sample size could have generated more accurate results. The importance of sample size is greater in quantitative studies compared to qualitative studies.

4. Lack of previous studies in the research area. Literature review is an important part of any research, because it helps to identify the scope of works that have been done so far in research area. Literature review findings are used as the foundation for the researcher to be built upon to achieve her research objectives.

However, there may be little, if any, prior research on your topic if you have focused on the most contemporary and evolving research problem or too narrow research problem. For example, if you have chosen to explore the role of

Bitcoins as the future currency, you may not be able to find tons of scholarly paper addressing the research problem, because Bitcoins are only a recent phenomenon.

5. Scope of discussions. You can include this point as a limitation of your research regardless of the choice of the research area. Because (most likely) you don't have many years of experience of conducting researches and producing academic papers of such a large size individually, the scope and depth of discussions in your paper is compromised in many levels compared to the works of experienced scholars.

You can discuss certain points from your research limitations as the suggestion for further research at conclusions chapter of your dissertation.

Question 12: Explain the limitation of research design.

Questions:

1. What do you mean by research design?

2. What is the concept of research design?

3. What are the important of research design?

4. Discuss the characteristics of research design.

5. Define exploratory research design and discuss the major types of qualitative techniques for exploratory research design.

6. Write advantage and disadvantage of exploratory research.

7. Define descriptive research design. Describe the types of descriptive research design.

8. Describe the characteristics, advantage and disadvantage of descriptive statistics.

9. Define experimental research design. Write the important of experimental research design. Also discuss the types of experimental research design.

10. Define quantitative and qualitative research.

11. Define and discuss case study method. Describe the different steps of case study approach.

12. Explain the limitation of case study.

13. Define dependent and independent variables.

14. Explain the limitation of research design.

